EDMS 02-307-1

TECNNICAL SPECIFICATION

FOR

LOW VOLTAGE METER PANEL

(SMART/PREPAID)

Issue: Feb. 2021 / Rev- 1

* Option items should be specified by …. EDC before tendering.
* Latest edition of EDMS 20-305 (Smart Meter) or EDMS 20-302 (Prepaid Meter) should be attached.
* Latest edition of EDMS 17-300 (Low voltage Current transformer) should be attached and current transformers ratio should be defined.

CONTENTS

1. **Scope**

This specification covers the minimum requirements for design, manufacturing and supply of three phase low voltage meters panels (Smart / Prepaid) to measure the electrical energy consumption on low voltage side for direct and indirect connection (0.4 kV / 50 Hz).

2. **Technical Specifications**

In addition to the requirements of this specification, the offered panel and its components should comply with the latest applicable standards as follows:

IEC 60439-1	Low Voltage Switchgear and Control gear General Assemblies.
IEC 60947-1	Low Voltage Switchgear and Control gear General Rules.
IEC 60947-2	Low Voltage Switchgear and Control gear for Circuit Breakers.
IEEE C37.13	Standard for Low-Voltage AC Power Circuit Breakers Used in Enclosures.
IEC 61076	Connectors for electronic equipment - Product requirements.

3. **Environmental Conditions**

Minimum ambient temperature	: -5°c.
Maximum ambient temperature	: 45°C (50 °C is optional).
Maximum relative humidity	: 95%.
Maximum altitude	: 1000 m.

4. **Panel Components**

4.1 Meter

The meter should comply with the latest edition of EDMS 20-305 (Smart Meter) or EDMS 20-302 (Prepaid Meter).

The meter should measure the electrical energy consumption via current transformers (xx/5) A, which should be approved by the Egyptian Electricity Holding Company (EEHC).

The meter should be installed so that it can be handled without having to open the inner door.

The voltage circuits of the meter should be connected directly to the incoming cables.

4.2 Main Circuit breaker (Motorized)

The main circuit breaker should comply with the latest edition of EDMS 19-303 (LV Circuit Breakers).

4.3 Busbars

All the bus-bars should be made from rectangular high-grade electrolytic copper with a purity of at least 99.999%, with a conductivity not less than 57 MS/m and current density not more than 1.5 A/mm^2. (refer to table 1)

The cross-section area of the neutral bus-bar should be equal to the cross-section area of the phases bus-bars.

The cross-section area of the earthing bus-bars should be at least 25% of the cross-section area of the phases bus-bars.

The main, neutral and earthing bus-bars should have holes and equipped with galvanized steel bolts and washers necessary for connecting cables.

All the bus-bars should be insulated with "heat-shrinkable" colored in red, yellow, blue, and black for the 3-phases, and neutral bus-bars, respectively.

The bus-bars should be mounted on insulators made from porcelain or cast resin according to the IEC standard.

4.4 Meter Control Circuit

A voltage summation circuit should be installed to ensure the continuity of supplying the circuit breaker’s motor when one or two phases fall and should be equipped with an electrical interlock to prevent short circuit incidence.

Control circuit should consist of the following:

1. Three auxiliary relays for disconnection, connection and alarm, they receive signals from the auxiliary contacts in the meter to disconnect and connect the circuit breaker.
2. Three miniature circuit breakers for the three phases.
3. One miniature circuit breaker for the alarm circuit.
4. One miniature circuit breaker connected to the input of the circuit breaker auxiliary contact.
5. Sound alarm circuit (the horn should be fixed to the body of the panel or the outer door), It disconnects after a time that is set by a timer while maintaining the degree of protection of the panel and should be provided with an indicator lamp and a reset button installed on the inner door, and the sound alarm circuit operates when the charging balance is about to end or the panel door is opened or reaching the value of the maximum load programmed on the meter.

6. A separate U-channel base for the control circuit components should be fixed by the inner door of the panel.

4.5 Current transformers

The panel should be provided with three current transformers according to the required capacity xx/5 A.

The current transformers should comply to the latest edition of EDMS 17-300 (Low voltage Current transformer).

5. Design criteria

1. The panel should be manufactured with dimensions suitable for the components as required by the size of the internal equipment and to allow for ease of installation and maintenance.
2. The panel body should be of galvanized sheet steel, cold drawn and chemically treated, with a thickness of not less than 2 mm (before painting) and electrostatically painted from the inside and the outside, including the interior and exterior doors.
3. The body of the panel should be supported by a frame of steel wickers with a thickness of not less than 2.5 mm.
4. The protection degree of the panel should not be less than IP54 (the door should be provided with inflatable gaskets) for sealing.
5. The roof of the panel should be sloping (the degree of inclination is not less than 6 degrees with the horizontal), if the panel is installed outside the building.
6. The panel should be supplied with four steel stirrups (welded to the body of the panel) of galvanized steel with a thickness of not less than 4 mm and painted with electrostatic paint with holes for fixing the panel to the wall.
7. The outer leaf and the inner door (inner mirror) must be ground with a tinned copper shield of at least 6 mm^2.
8. A grounding bolt should be installed on one side of the panel for panel grounding.
9. The cable entry and exit holes should be at the panel bottom and be suitable for the cable cross-section area and equipped with PVC glands suitable for the cable section area (according to the rated load of the panel), with degree of protection not less than IP54.

10. The outer door of the panel should be locked by a nickel-plated ball locker with a blot to seal the panel.
11. The inner door of the panel should have three high-efficiency LED voltage indicating lamps and two other LED lights to indicate the disconnection and connection status of the circuit breaker and two LED lights to indicate the operation of the panel (manual / automatic), all should be equipped with fuses.
12. The inner door (hinged mirror) should be equipped with a nickel-plated ball locker with a blot and a protrusion for the possibility of installing a lock on it.
13. The possibility of agglomerating the inner door should be provided.
14. The panel should be equipped with a limit switch installed on a fixed base that cannot be moved so that the panel inner door presses when closed.
15. The limit switch should be controlled by the meter.
16. The inner door of the plate should have an opened area that enables to deal with the meter without having to open it.

6. Panel operating

The operating mode of the panel is the automatic mode, and the manual mode switching should be done only in exceptional cases by the Electricity Distribution Company only and it should be temporary until the causes for disconnection are cleared and the necessary maintenance is done or the power is returned in the event of a disconnection of the smart meter (the manual operation mode can be completely canceled according to Electricity Distribution Company requirements).

The power should be cut off by the circuit breaker, and it should not be reenergized except by the Electricity Distribution Company in the following cases:

1. Opening the inner door of the panel.
2. Expiry of the balance in the case of prepayment or the expiration of the grace period for deferred payment.
3. Attempt to manipulate the opening of the meter cover or the terminal cover.
4. The customer’s load exceeds the maximum load programmed on the meter and the number of times programmed in the meter.

7. Nameplate

1. the nameplate should be metal, and should be riveted to the panel.
2. The data should be written in a manner that cannot be erased (engraving or laser print).
3. The nameplate should be of suitable dimensions, and it should include the following data:
   i. Electricity Distribution Company name.
   ii. Current transformer ratio (xx/5) A
   iii. A printed warning in Arabic " ممنوع فتح الباب الداخلي للوحة، ومن يخالف ذلك يتعرض للغرامة "

8. Inspection and Testing

1. Electricity Distribution Company (…EDC) reserves the right to witness the routine tests of the panel.
2. Unless otherwise specified or approved in writing by …EDC, all panels should be tested at the supplier factory or in an Egyptian approved laboratory which accepted by the distribution company at the supplier's expense in accordance with the latest applicable issues of IEC standard formerly mentioned.
3. Routine tests should be carried out on the panel complete with the meter is built in it and in accordance with the requirements of this specification.
4. The tenderer should provide all test certificates and acceptance certificates.
5. Attendance of ...EDC representative during fabrication (assembly) stages, and testing should not relieve the supplier of his full responsibility for furnishing the panel in accordance of this specification requirements, and should not give him the right to invalidate any claim which …EDC, may make because of defective or unsatisfactory material or faulty workmanship.
6. The type test certificates should be submitted with offer to...EDC.
7. The tenderer should submit a quality control procedure, and an approval certificate from the Egyptian Electricity Holding Company (EEHC) for the panel and its basic components with his offer.

9. Drawings and Catalogues

The tenderer should submit, along with his technical offer, a complete copy of the technical catalogs and guarantee tables (stamped and signed), as well as, a copy of the factory's quality control procedures. The bid that does not include the catalogs and guarantee tables shall be rejected.

The tenderer should submit the operating manuals and the Wiring diagrams.

10. Warranty

The warranty period should be one year from the date of operation, with a maximum of 18 months from the date of delivery.

Table (1)

ITEM	Specifications										
Panel Current (A)	**100**	**150**	**200**	**300**	**400**	**500**	**600**	**800**	**1000**	**1500**	**2500**
Cross section area of main bus bars (mm^2)	100	125	200	250	400	400	500	800	1000	1200	2000
Three current transformers, ring type class 0.5	100/5	150/5	200/5	300/5	400/5	500/5	600/5	800/5	1000/5	1500/5	2500/5
Circuit breaker type:	MCCB							ACB			
Frame current at 50°C ambient temperature, A	125	200	250	400	500	630	800	1000	1600	2000	4000
Nominal breaking at 400V, Icu = Ics, KA	10	35	35	35	35	35	50	50	50	65	85
Nominal current at 50 °C In, A	125	200	250	400	500	630	800	1000	1250	2000	3200
Thermal release setting at 50°C, A	(0.8-1) In			(0.4-1) In							
Magnetic release setting range, A	(5-10) In			(2-10) In							
Rated insulation voltage, (Ui)V	690	750									
Rated impulse withstands voltage (Uimp) kV	6	8									

Table (2)

Reference Ampere Rating at 40°C (104° F)	Ampere Rating at:		
	25°C (77° F)	50°C (122° F)	60°C (140° F)
15	17	13	11
20	22	18	16
25	28	23	21
30	33	28	26
35	39	30	25
40	44	37	34
50	55	46	42
60	66	56	52
70	77	65	60
90	99	84	78
100	110	94	87
125	137	114	100
150	165	136	120
175	192	159	140
200	220	182	160
225	247	205	180
250	275	235	220
300	330	276	252
350	385	325	301
400	440	372	340
500	550	468	435
600	660	564	525
700	770	658	613
800	880	754	704
900	990	828	749
1000	1100	900	825
1200	1320	1090	1000
1400	1540	1304	1148
1600	1760	1500	1320
1800	1980	1690	1485
2000	2200	1880	1650
3200	3520	3000	2650
4000	4400	3760	3300
5000	5500	4700	4125

Guarantee Table (1)
Enclosure and Bus-Bars

1- Enclosure: -

- Manufacturer name :-----------
- Type :-----------
- Material :----------
- Thickness of steel sheet before painting :---------- mm
- Dimensions :----------
  * Width : -------- mm
  * Depth : -------- mm
  * Height : --------- mm

2-Bus-bars: -

- Material : ---------
- Cross section
  * Per phase : -------- mm^2
  * Neutral bus : --------- mm^2
  * Earth bus : --------- mm^2
- Purity of copper : ---------- %
- Conductivity of copper : ---------- SM/m
- Minimum clearance between phases : ---------- mm
- Minimum clearance to earth : ---------- mm
- Continuous current inside the enclosure
  * At 50°C for indoor :------------ A
- Dynamic withstand current, peak : ----------- kA
- One- second thermal withstand current : ----------- kA
- Rated insulation voltage : ----------- V
- One- second power frequency withstand voltage : ----------- V

I / We guarantee the technical data given above for the offered equipment.

Signature: -----------------

Date: -----------------

Guarantee Table (2)
Main Circuit - Breaker

- Manufacturer name :-----------
- Standard specification :----------
- Type :----------
- Rated current inside the enclosure
  - * At 50˚C :--------- A
- Rated insulation voltage : --------- V
- Rated breaking capacity at 400V : --------- kA
- Rated making capacity, peak : ---------- kA
- Rated current of thermal release inside the enclosure (Ir):
  - * At 50˚C :----------- A
- Thermal release setting value at 50˚C :---------- A
- Short circuit release setting value : ---------- A
- One-second power frequency withstand voltage : ---------- V
- Material of main contacts : ---------
- Material of arcing contacts : -----------
- Type of arc control device : ---------
- Material of arc control device : --------

I / We guarantee the technical data given above for the offered equipment.

Signature: -----------------

Date: -----------------